

Andrei Șerban

Senior QA Automation Engineer · Test Architect

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Senior QA Automation Engineer & Test Architect with two concurrent senior engagements. At TOKERO (European crypto exchange) I am sole architect of the QA stack — Playwright (.NET / C#), NBomber, and a custom Blazor reporting platform. At Heaven Solutions I lead automation for the Deutsche Bahn SAP ERP Integrated Railway Management System — Karate API automation suite and Playwright UI framework. Across both engagements: 5 production QA frameworks, 55% automated coverage, ~60% faster feedback loops, and €2M+ in prevented system failures. Active practitioner of AI-augmented QA: Claude Code skills, MCP integrations (Playwright, Supabase), and spec-driven automation. I mentor junior QA engineers and treat quality as a shared responsibility — not a gate.

CORE COMPETENCIES

Testing Frameworks Playwright (TypeScript & .NET), Selenium WebDriver, Karate API, NBomber, TestNG, JUnit, xUnit / NUnit, Cucumber / BDD

Languages TypeScript, JavaScript, Java, C# / .NET Core, SQL

AI-Augmented QA Claude Code, Custom Skill Authoring (.claude/skills), MCP Integration (Playwright, Supabase), Subagent-Driven Test Development, AI-Assisted Test Generation & RCA, Prompt Engineering for QA, Spec-Driven / Plan-First Automation

Platforms & DevOps Docker, Kubernetes, GitLab CI/CD, Jenkins, Azure DevOps, OpenLens, Grafana, Kibana, Prometheus

API & Integration Postman, ReadyAPI, SoapUI, Swagger, SignalR / WebSocket testing

Test Management Jira, XRay, Confluence, Polarion, Bugzilla / Deskzilla

Databases MariaDB, PostgreSQL, Oracle, SQL query authoring for back-end validation

Methodologies Agile / Scrum (PSM I), BDD, Shift-Left Testing, Risk-Based Testing, UAT facilitation, Root Cause Analysis

PROFESSIONAL EXPERIENCE

Senior QA Automation Engineer & Test Architect · TOKERO Crypto Exchange Jul 2025 – Present

Concurrent senior engagement · European cryptocurrency exchange · 150+ coins

- **Sole architect & owner of 3 production QA systems:** the **Playwright (.NET / C#) functional framework** in production since 2025, plus the **NBomber performance suite** and a custom **Blazor reporting & dashboard platform** — both shipped to production in 2026.
- **77 Page Objects, 49 test classes, 11 performance scenarios, and 27 performance profiles** owned and maintained end-to-end across the QA estate.
- Built the in-house **pulse** reporting platform from scratch (C#, .NET 10, Blazor, EF Core) with candlestick + trend analytics, flaky-test detection, and a 10% regression threshold — gave product, ops, and compliance stakeholders live visibility into release readiness.
- Authored **5+ custom Claude Code skills** now standard for test and Page Object generation across the QA team; integrated Playwright MCP for agentic UI verification — first AI-augmented test authoring workflow at TOKERO.

- Designed test architecture for high-throughput trading flows, KYC / onboarding paths, and multi-market deposit / withdrawal logic across 27 jurisdictions.
- Built the first real Blazor SignalR independent-circuit perf scenario in the codebase — closed a load-test gap that NBomber didn't cover out of the box.
- *Stack: Playwright .NET, C#, .NET 10, Blazor, MudBlazor, ApexCharts, NBomber, EF Core, PostgreSQL (Supabase), Azure Key Vault, GitLab CI/CD, Claude Code, Playwright MCP, Git.*

QA Automation Engineer · Heaven Solutions · Deutsche Bahn (SAP ERP IRMS) Jan 2023 – Present

Iasi, Romania · SAP ERP Integrated Railway Management System for Europe's largest transportation network

- **Architect & owner** of the QA automation stack for the Deutsche Bahn SAP ERP Integrated Railway Management System: **Karate-based API regression suite (BDD, Java)** and **Playwright UI automation framework (TypeScript)**. **55% automated coverage** protecting **500+ critical workflows**, aligned with Germany's €40B transportation modernization program.
- Prevented **€2M+** in potential system failures through proactive defect detection; reduced customer-impacting defects by **25%** and slashed critical issue resolution time by **50%** via advanced monitoring and rapid diagnosis.
- **40% faster testing cycles** and **35% faster releases** by replacing legacy Selenium with Playwright; slashed manual ERP testing by **85%**.
- Leading the automation team; led 15+ engineers in critical system bug resolution and **mentored 5+ junior QA engineers** on test design, framework architecture, and code review.
- Built E2E Selenium / Java / Cucumber suites for legacy modules; integrated everything into **GitLab CI/CD** with parallelized execution — cut feedback loops by **~60%** on long suites.
- Live-debug production issues using Kibana, Grafana, and OpenLens against the Kubernetes cluster; author SQL validation queries against MariaDB to confirm developer implementations.
- *Stack: Karate, Playwright, TypeScript, Java, Angular, Java Spring Boot, SAP GUI, Postman, Selenium, Cucumber, JUnit, MariaDB, SQL, Git, Kubernetes, Docker, OpenLens, Grafana, Kibana, Swagger, XRay, Jira, Confluence.*

Software Tester (on-site, Germany) · Heaven Solutions · DentsplySirona May 2022 – Dec 2022

Bensheim, Hessen, Germany · Medical device CAD/CAM software & hardware testing

- Led precision testing for medical manufacturing systems with $\pm 0.001\text{mm}$ accuracy requirements; 100% on-time deliverables.
- Resolved 100+ critical defects with high-clarity bug reports, reducing developer resolution time by **~40%** on CAD/CAM workflows.
- Acted as interim team lead and primary point of contact for the client during the incumbent's absence.
- Acceptance, regression, sanity, and functional/performance testing on both software and hardware; Linux-server checks on Storage Hubs; SQL back-end validation.
- Participated in Root Cause Analysis sessions; supported the test manager on test plans, test points, and TPFs; built executive-facing reports.
- *Tools: CAD/CAM software & hardware, SQL, PostgreSQL, Oracle, GitLab, GitHub, Postman, Polarion, Bugzilla / Desktzilla, Jira, MS Office.*

Junior Web Developer · Display Events Agency · Happy Media May 2021 – May 2022

Iasi, Romania · Full-service advertising agency, 200+ SMEs, 2000+ campaigns across Eastern Europe

- Delivered web platforms that helped raise client acquisition by 35% and reduce manual work by 80% for the agency's campaign operations.
- Maintained 99.9% uptime across SME campaign sites; authored installation, execution, and maintenance documentation; tested every release before QA hand-off.

- Active participant in team brainstorming, scoping, and post-mortems.
- *Stack: HTML, CSS, JavaScript, Angular, Java, WordPress, Elementor, cPanel, PHP.*

Intern — Junior System Administrator · RLogicDesign

Jun 2017 – Sep 2017

Bacău, Romania

- Maintained Windows and network systems; software deployment, install, and troubleshooting for printers and end-user devices.

FEATURED PROJECTS

Architecture overviews, design patterns, framework structure, and AI-augmented workflows are documented in detail on my portfolio at aserban.ro (see Architecture & Approach section).

TOKERO QA Automation Platform · Ongoing · Sole architect & owner · Senior QA / Test Architect role at TOKERO

- End-to-end automation stack for a European crypto exchange: Playwright (.NET / C#) functional framework — **77 Page Objects, 49 test classes** (in production since 2025); NBomber performance suite — **11 scenarios, 27 profiles**, incl. first real Blazor SignalR perf scenario; and a custom Blazor reporting platform (C#, .NET 10) — both shipped to production in 2026.

Deutsche Bahn — SAP ERP Integrated Railway Management System QA Automation Framework · Ongoing ·

Architect & owner · Heaven Solutions client engagement

- Sole architect & owner of the QA automation stack supporting Europe's largest transportation network: Karate-based API regression suite (BDD, Java) and Playwright UI automation framework (TypeScript). **€2M+ prevented system failures, 55% automated coverage, 40% faster testing cycles, 35% faster releases, and 85% reduction in manual ERP testing** — aligned with Germany's €40B transportation modernization program. Stack: Karate, Playwright, Angular, Java Spring Boot, SAP GUI.

DentsplySirona — Medical Device CAD/CAM Testing · Delivered

- Precision testing for medical manufacturing systems with $\pm 0.001\text{mm}$ tolerance; 100% on-time deliverables as interim team lead during critical product launches.

Happy Media — Digital Campaign Management Platform · Delivered

- Built and maintained a campaign platform serving 200+ SMEs across Eastern Europe; 99.9% uptime, 35% lift in client acquisition rates, 80% reduction in manual ops work.

AI-AUGMENTED QA PRACTICE

Active practitioner and advocate for AI in software quality. Current work and obsessions:

- **Custom Claude Code skill authoring** (.claude/skills) — encoding team standards into reusable, repo-aware skills that scale across QA engineers.
- **MCP integrations** — Playwright MCP for agentic UI testing; Supabase MCP for data-aware regression flows.
- **Subagent-driven test development** — orchestrating specialist agents (planner, generator, reviewer) for higher-quality test artifacts.
- **AI-assisted test generation, debugging & RCA** — using LLMs to draft failing tests from specs and to triage flaky runs faster.
- **Plan-first / spec-driven automation** — flipping the order: specs → executable plans → implementation, so tests stay aligned with intent.
- **Visual-regression workflows & Playwright MCP** — current build-out for higher-confidence UI parity at scale.

EDUCATION & CERTIFICATIONS

Engineer, Electronics & Automation · "Gheorghe Asachi" Technical University of Iași 2015 – 2020
MATLAB, Arduino, Assembler, MAX+PLUS II, Java SE, Networking, FEMM (electromagnetic / magnetic fields), team operations.

General Certificate of Highschool · Colegiul Național "Vasile Alecsandri", Bacău 2011 – 2015
Computer science track — C++, algorithms, Office, team work.

Certifications

- **ISTQB® Certified Tester Foundation Level (CTFL)** — Brightest, 2022. *Black/white-box, test design, test management, risk & maintenance testing.*
- **Professional Scrum Master™ I (PSM I)** — Scrum.org, 2023. *Empiricism, Scrum values, self-managing teams, product agility.*
- **Google Developer Challenge Scholarship** — Basic Android (Udacity, sponsored by Google).

Continuous Learning

- 12+ specialized testing & engineering courses completed.
- Active: Playwright at expert level (3+ years hands-on), GitLab CI/CD daily, container technologies in production, API testing core competency.
- Currently exploring: AI testing tools and visual regression testing; advancing toward Tech / QA Lead via strategic test leadership, mentoring, and enterprise-scale automation frameworks.

LANGUAGES

- **Romanian** — Native.
- **English** — C1 (Listening, Reading, Spoken interaction); B2 (Spoken production, Writing).
- **French** — B2 (Listening, Reading); B1 (Speaking, Writing).

BEYOND TECH

Parallel career in human movement and recovery, ongoing since 2018:

- **Massage Therapist** (self-employed, 2019 – June 2024) — therapeutic, cupping, scraping, fascial release.
- **Physiotherapist** — National Individual Championships U18, Concord Service Center (2020).
- **Personal Trainer & Fitness Instructor** — Young Gym, Motivation Gym, Vivertine (2018 – 2022). Accredited PT Level 1, Fitness Scandinavia.
- **Specialized training** — Foundational MAT / Dynamic Lower Body (Erik Dalton), Heavenly Head Massage (Thai Healing Massage Academy), Human Optimization (Functional Patterns by Naudi Aguilar), Technician Masseur Supramodul I & II (final grade 10).

This dual focus shapes how I approach engineering: with patience, attention to body language in meetings, and the conviction that systems — software or human — perform best when well-aligned and given space to recover.

OTHER

Driving licences AM · B1 · B

Date of birth 31 May 1996 · **Nationality** Romanian